Discovering and Mitigating Visual Biases through Keyword Explanation

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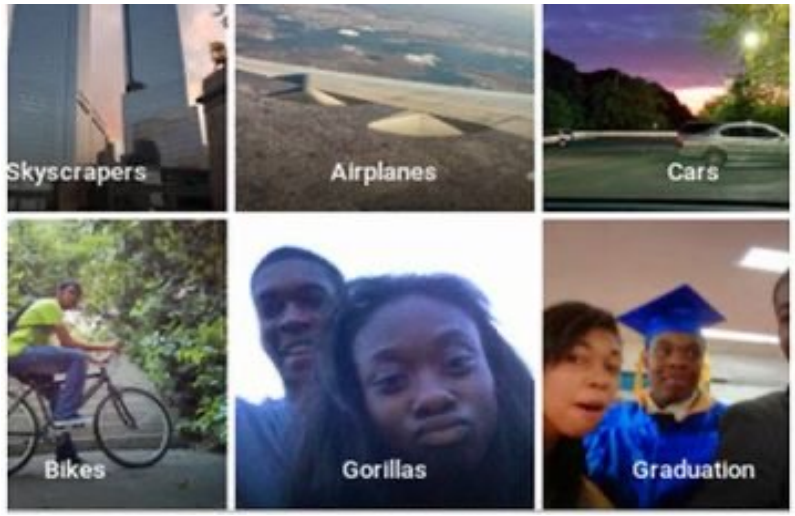
CVPR 2024 Highlight

2024.07.19.

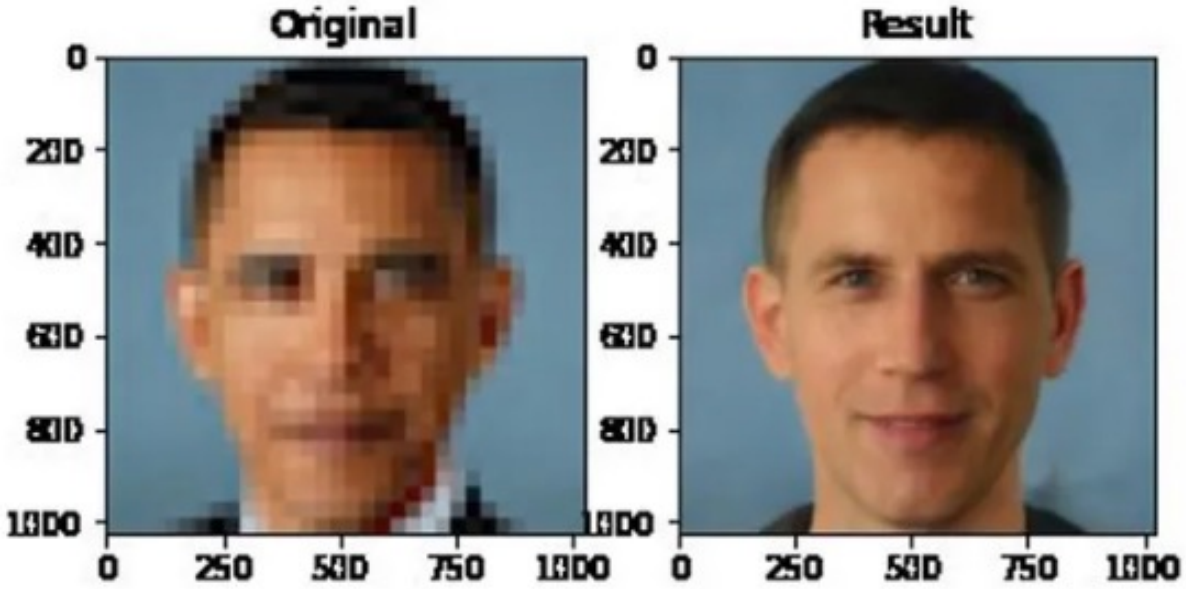
https://arxiv.org/pdf/2301.11104

Biases are everywhere in ML domain

- There exist visual biases inherited from ML algorithm in real-world application



Google Photos automatic tagging



PULSE algorithm: low pixel image to high resolution image

https://www.bbc.com/news/technology-33347866
https://www.theverge.com/21298762/face-depixelizer-ai-machine-learning-tool-pulse-stylegan-obama-bias

These visual biases pose several critical problems

- Biases may cause fairness issue
- Biases may harm model performance



Classifier **mispredicts** blond male !

Rare in training examples (bias)

[Sagawa et al., 2019] Distributionally Robust Neural Networks for Group Shifts

However, visual biases are not interpretable

- Prior works visualized spurious features that are not human-readable
- Thus, they are hard to be directly utilized for debiasing



(a) **class:** band aid, **spurious feature:** fingers, **-41.54%**

(b) **class:** space bar, **spurious feature:** keys, **-46.15%**

(c) **class:** plate, **spurious feature:** food, **-32.31%**

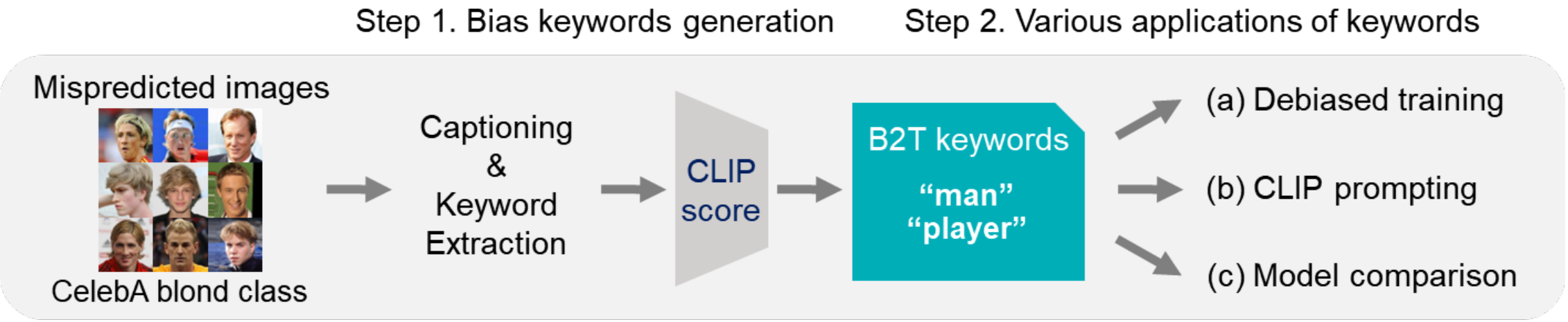
(d) **class:** butterfly, **spurious feature:** flowers, **-21.54%**

(e) **class:** potter's wheel, **spurious feature:** vase, **-21.54%**

[Singla et al., 2022] Salient ImageNet: How to Discover Spurious Features in Deep Learning

B2T: Bias-to-text

- We use language to interpret visual biases
- We first **extract** B2T keywords, then use them to **various applications**: (a) debiased training, (b) CLIP prompting, and (c) model comparison



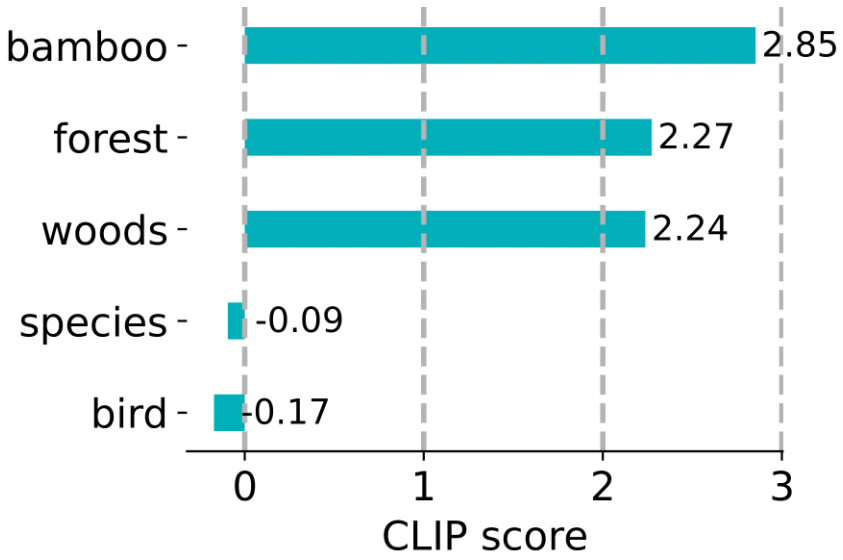
CLIP score

- CLIP score measures the similarity between keyword a and correctly or incorrectly classified images x from a validation set $\mathcal{D}$

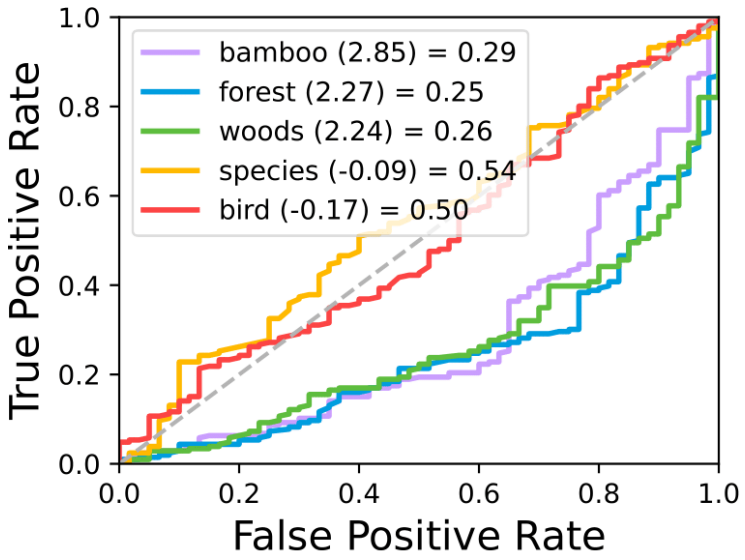
$$s_{\text{CLIP}}(a; \mathcal{D}) := \text{sim}(a, \mathcal{D}_{\text{wrong}}) - \text{sim}(a, \mathcal{D}_{\text{correct}}).$$

Validation of the CLIP score

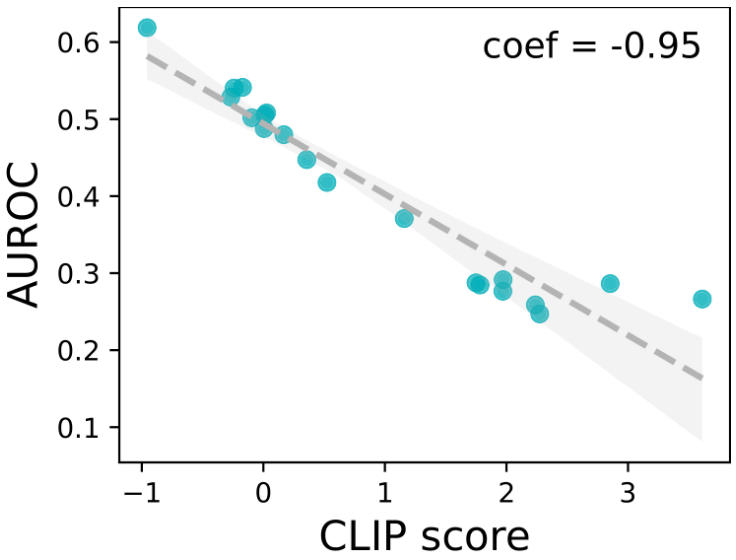
- CLIP score effectively identifies incorrect bias keywords
- e.g.) waterbird class in the Waterbirds dataset



(a)CLIP score



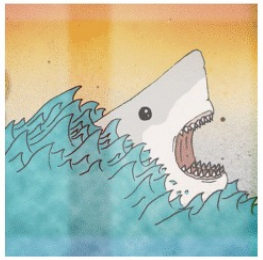
(b) ROC curve of subgroup acc.



(c) Correlation of CLIP score and AUROC

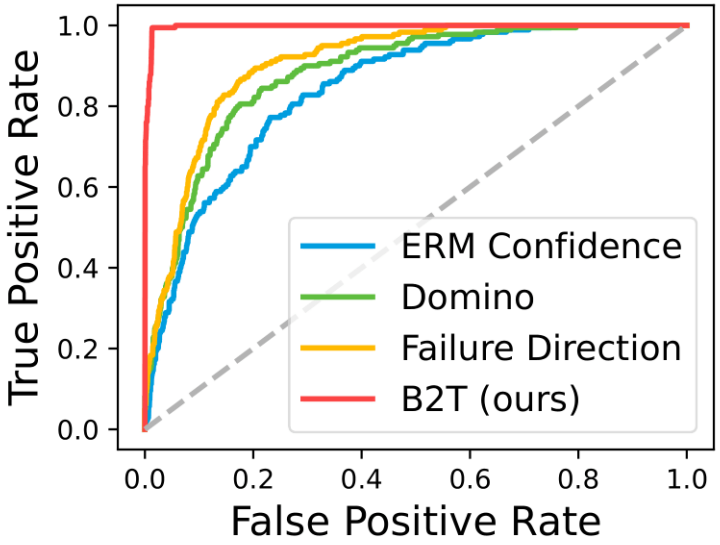
Can B2T identify the known biases?

- B2T discovers **spurious correlations** and **distributions shifts**
- e.g.) “man” for CelebA blond / “forest” and “ocean” for Waterbirds
  “illustration” and “drawing” for IN-R / “snow” and “window” for IN-C

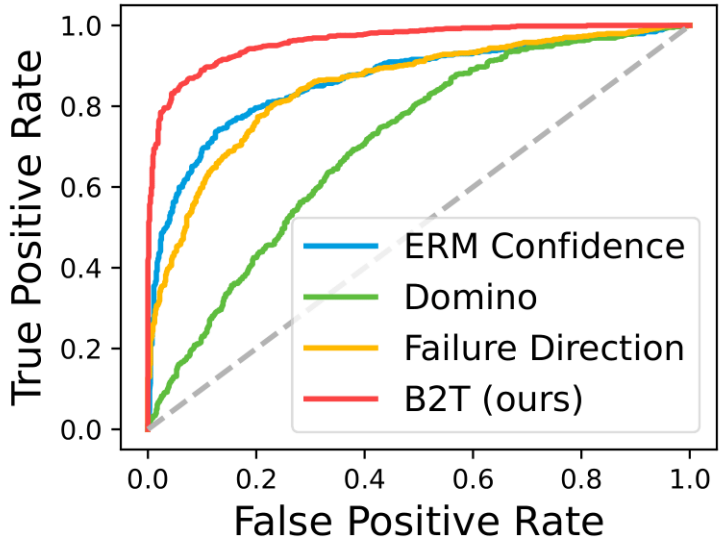
	(a) CelebA blond		(b) Waterbirds		(c) ImageNet-R		(d) ImageNet-C snow / frost	
Keyword	Man		Forest	Ocean	Illustration	Drawing	Snow	Window
Samples								
Actual	blond	blond	waterbird	landbird	backpack	white shark	airliner	American egret
Pred.	not blond	not blond	landbird	waterbird	maze	envelope	damselfly	quill
Caption	person, a **man** with a beard.	actor as a young **man**.	a bird in the **forest**.	a bird in the **ocean**.	hand drawn **illustration** of a backpack.	a **drawing** of a shark attacking […]	airliner in the **snow**, photo.	a bird on a frozen **window**.

Sample-wise bias labeling

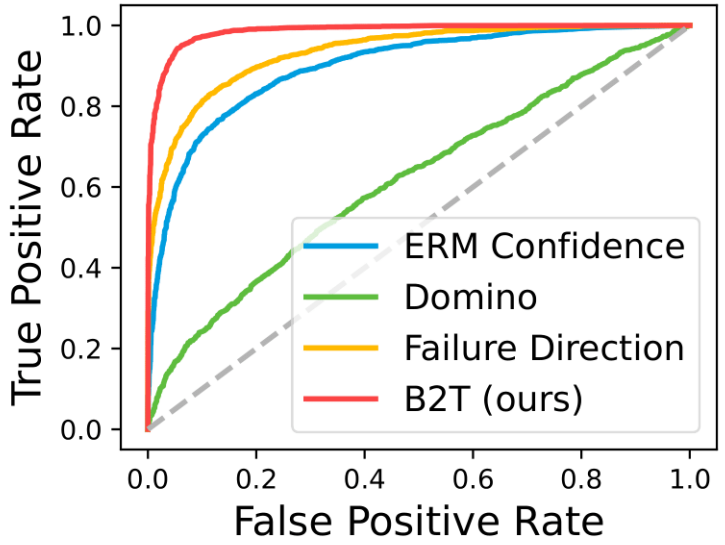
- B2T successfully infers **sample-wise bias (or group) labels**



(a) CelebA blond



(b) Waterbirds waterbird



(c) Waterbirds landbird

Novel real-world biases

- B2T explores **novel biases** in larger datasets
- e.g.) “cave,” “fire,” “bucket,” and “hole” for Dollar Street
  “flower,” “playground,” “baby,” and “interior” for ImageNet

	(e) Dollar Street				(f) ImageNet			
Keyword	Cave	Fire	Bucket	Hole	Flower	Playground	Baby	Interior
Samples								
Actual	wardrobe	stove	plate rack	toilet seat	ant	horizontal bar	stethoscope	monastery
Pred.	poncho	caldron	oil filter	wheelbarrow	bee	swing	baby pacifier	arched ceiling
Caption	the **cave** is full of surprises.	a **fire** in the kitchen.	a **bucket** of water and a few tools.	the **hole** in the ground.	a yellow **flower** with a black head.	person on a swing in the **playground**.	a newborn **baby** boy in a stethoscope.	the **interior** of the church.

Debiased DRO training

- Bias keywords can be used as group names for **debiased distributionally robust optimization (DRO) training**

Method	GT	CelebA blond		Waterbirds	
		Worst	Avg.	Worst	Avg.
ERM	-	47.7±2.1	94.9	62.6±0.3	97.3
LfF [55]	-	77.2	85.1	78.0	91.2
GEORGE [74]	-	54.9±1.9	94.6	76.2±2.0	95.7
JTT [44]	-	81.5±1.7	88.1	83.8±1.2	89.3
CNC [86]	-	88.8±0.9	89.9	88.5±0.3	90.9
DRO-B2T (ours)	-	**90.4**±0.9	93.2	**90.7**±0.3	92.1
DRO [66]	✓	90.0±1.5	93.3	89.9±1.3	91.5

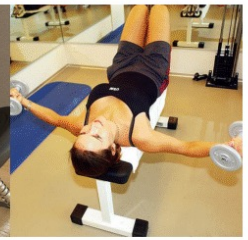
CLIP zero-shot prompting

- Bias keywords can **improve the CLIP zero-shot classifier** by integrating them into prompt

	CelebA blond		Waterbirds	
	Worst	Avg.	Worst	Avg.
CLIP zero-shot	76.2	85.2	50.3	72.7
+ Group prompt [85]	76.7	87.0	53.7	78.0
+ B2T-neg prompt	72.9	88.0	45.4	70.8
+ B2T-pos prompt (ours)	**80.0**	87.2	**61.7**	76.9

Model comparison

- Bias keywords can be used to **analyze and compare different classifiers** based on their keywords
- e.g.) architecture: ResNet vs. ViT

Keyword	Work		Supermarket	
Samples				
ViT-B	O	O	O	O
RN50	O	X	O	X
Actual (RN50)	dumbbell	dumbbell	shopping basket	shopping basket
Pred (RN50)	dumbbell	horizontal bar	shopping basket	grocery store
Caption	a set of dumbbells with weights.	person **works** out in the gym.	a basket full of food.	woman shopping in a **supermarket**.

Label diagnosis

- B2T can diagnose **common labeling errors**, such as mislabeling and label ambiguities

Keyword	Bee	Boar	Desk	Market
Samples				
Label	fly	pig	computer mouse	custard apple
Pred.	bee	wild boar	desktop computer	grocery store
Caption	a **bee** on a yellow flower.	wild **boar** in the forest.	the **desk** in the office.	fruit and vegetables at the **market**.

B2T: Bias-to-Text

- We interpret visual biases as **keywords** that enables the **discovery of novel biases** and the **effective debiasing of vision models**

